

Privacy-Preserving Machine Learning with Federated Learning and Differential Privacy: A Comprehensive Survey

Abstract

The proliferation of data-driven artificial intelligence has precipitated a critical tension between the utility of large-scale machine learning models and the imperative to protect user privacy. Federated Learning (FL) emerged as a paradigmatic shift, enabling collaborative model training across decentralized data silos without exchanging raw samples. However, the transmission of model updates—gradients or weights—remains vulnerable to sophisticated inference attacks, including gradient leakage, membership inference, and model inversion. Differential Privacy (DP) has consequently become the gold standard for providing rigorous, mathematical guarantees against such privacy breaches. This survey provides a comprehensive synthesis of the intersection between Federated Learning and Differential Privacy (DP-FL), analyzing over 300 recent academic contributions. We construct a unified taxonomy categorizing DP-FL methodologies based on trust models (Centralized, Local, Distributed, and Shuffle), privacy granularity (sample-level vs. client-level), and optimization strategies. We critically examine methodological innovations ranging from adaptive noise injection and gradient sparsification to hybrid cryptographic-DP frameworks and wireless over-the-air computation. Furthermore, we explore the intricate trade-offs between privacy, utility, fairness, and communication efficiency, highlighting how recent advances in privacy accounting, robust aggregation, and personalized learning mitigate the inherent utility degradation of DP mechanisms. Finally, we discuss open challenges in deployment, including the vulnerability of secure aggregation to Sybil attacks, the computational overhead of hybrid protocols, and the nascent field of empirical privacy auditing.

1. Introduction

The era of big data has fundamentally transformed machine learning, yet it has simultaneously exacerbated concerns regarding data sovereignty and individual privacy. Traditional centralized learning requires aggregating sensitive data into a single repository, creating a high-value target for breaches and violating regulations such as the GDPR and CCPA. Federated Learning (FL) addresses this by decentralizing the training process: clients (e.g., mobile devices, hospitals, financial institutions) train local models on private data and share only model updates with a central server for aggregation [1], [2]. While FL prevents direct access to raw data, it does not inherently guarantee privacy. Seminal works have demonstrated that adversaries, including honest-but-curious servers, can reconstruct training data from shared gradients via Deep Leakage from Gradients (DLG) attacks or infer membership in the training set [3], [4], [5].

To mitigate these risks, Differential Privacy (DP) has been integrated into FL workflows. DP ensures that the output of an algorithm is statistically indistinguishable whether or not any single individual’s data is included in the dataset, thereby bounding the information leakage about any specific participant [6], [7]. The integration of DP into FL, however, introduces complex challenges. The addition of calibrated noise to mask individual contributions inevitably degrades model utility, a phenomenon known as the privacy-utility trade-off [8], [9]. Moreover, the distributed nature of FL complicates privacy accounting, as the composition of privacy loss over multiple communication rounds and heterogeneous client participation must be rigorously tracked [10], [11].

Recent literature has exploded with diverse approaches to reconcile these competing objectives. Methodologies have evolved from simple Local Differential Privacy (LDP), where clients add noise

locally before transmission, to more sophisticated Distributed Differential Privacy (DDP) and Shuffle models that leverage secure aggregation and privacy amplification by shuffling to achieve central-level utility with local-level trust assumptions [12], [13], [14]. Innovations now span adaptive clipping strategies [15], [16], communication-efficient quantization mechanisms that double as privacy filters [17], [18], and hybrid frameworks combining DP with Homomorphic Encryption (HE) or Secure Multi-Party Computation (MPC) for defense-in-depth [19], [20]. Furthermore, the field has expanded to address fairness [21], [22], robustness against Byzantine attacks [23], [24], and application-specific constraints in domains such as healthcare [25], [26], finance [27], and wireless communications [28], [29].

This survey synthesizes the state-of-the-art in Privacy-Preserving Machine Learning (PPML) at the intersection of FL and DP. We organize the field into a coherent taxonomy, analyzing the theoretical foundations, algorithmic innovations, and empirical evaluations of over 300 distinct studies. We begin by establishing a taxonomy of trust models and privacy granularities. We then delve into core methodological families, including noise injection mechanisms, communication-efficient privatization, adaptive optimization, and hybrid cryptographic approaches. Subsequently, we examine specialized applications in vertical federated learning, decentralized topologies, and resource-constrained environments. Finally, we discuss critical challenges, including the limitations of current privacy accounting, the vulnerability of secure aggregation to active attacks, and the emerging need for empirical privacy auditing, before outlining future research directions.

2. Method

The integration of Differential Privacy into Federated Learning is not a monolithic technique but a diverse ecosystem of protocols tailored to specific threat models, network topologies, and utility requirements. We categorize the methodological landscape based on the locus of trust, the granularity of privacy protection, and the mechanism of noise injection or perturbation.

2.1 Taxonomy of Trust Models and Privacy Granularity

A fundamental distinction in DP-FL lies in the trust assumption regarding the aggregator and the resulting privacy model.

2.1.1 Centralized vs. Local vs. Distributed Differential Privacy In **Centralized Differential Privacy (CDP)**, a trusted curator (server) aggregates raw or un-noised updates and adds noise to the final aggregate. While this offers the best utility-privacy trade-off, it requires trusting the server not to inspect individual updates [8], [30]. In contrast, **Local Differential Privacy (LDP)** assumes an untrusted server; clients perturb their data or gradients locally before transmission. While LDP provides strong privacy guarantees, it suffers from significant utility degradation due to the accumulation of noise from every participant [31], [32].

Distributed Differential Privacy (DDP) emerges as an intermediate paradigm. By combining LDP with Secure Aggregation (SecAgg), DDP allows clients to add noise locally, but the server only sees the sum. If the noise is coordinated or if the aggregation protocol hides individual contributions, the system can achieve privacy amplification, effectively approaching CDP guarantees without a trusted server [33], [34]. Recent surveys emphasize that the choice between these models depends heavily on the “protection object”—whether protecting individual samples (sample-level DP) or entire client datasets (client-level DP) [8], [35]. For instance, client-level DP is crucial in cross-device FL where a single user contributes many samples, whereas sample-level DP is standard in

cross-silo settings [35], [36].

2.1.2 The Shuffle Model and Privacy Amplification The **Shuffle Model** represents a significant architectural innovation, introducing a trusted shuffler between clients and the analyzer. Clients send encrypted, locally perturbed messages to the shuffler, which randomly permutes them before forwarding to the analyzer. This process breaks the link between the sender and the message, providing “privacy amplification by shuffling” [12], [37]. Papers such as [12] (FLAME) and [13] (RAIN) demonstrate that shuffling allows clients to use weaker local noise while achieving strong central privacy guarantees. FLAME, for example, employs a double amplification mechanism combining shuffling with subsampling and dummy padding to reduce noise injection significantly [12]. Similarly, RAIN utilizes a two-server secret-shared shuffle to enable robust aggregation in the presence of malicious actors while maintaining the privacy benefits of the shuffle model [13]. Theoretical analyses confirm that the shuffle model can bridge the gap between the high utility of CDP and the strong trust assumptions of LDP, provided the shuffler does not collude with the analyzer [14], [38].

2.2 Noise Injection and Perturbation Mechanisms

The core of DP-FL is the mechanism used to inject noise or perturb updates to satisfy privacy bounds.

2.2.1 Gradient Clipping and Gaussian/Laplace Mechanisms Standard DP-SGD involves clipping per-sample gradients to bound sensitivity, followed by the addition of Gaussian or Laplace noise [39], [9]. In FL, this is adapted to clip client updates. However, fixed clipping bounds often lead to excessive bias or insufficient privacy. Adaptive clipping strategies have gained prominence. **Adap DP-FL** dynamically calculates clipping thresholds based on noisy historical gradient norms and validation loss trends, significantly improving utility over static baselines [15]. Similarly, **DP-FedSAM** integrates Sharpness Aware Minimization (SAM) to seek flat minima, which naturally reduces the L_2 norm of updates, thereby mitigating the information loss caused by clipping and enhancing robustness to DP noise [16].

The choice of noise distribution is also critical. While Gaussian noise is standard for (ϵ, δ) -DP, discrete mechanisms are essential for secure aggregation over finite fields. The **Skellam Mixture Mechanism (SMM)** injects noise drawn from a mixture of shifted Skellam distributions, allowing direct perturbation of real-valued gradients while maintaining integer outputs compatible with MPC, outperforming standard discrete Gaussian mechanisms in communication-constrained settings [40]. **QMGeo** takes a different approach, using stochastic quantization with a mixed truncated geometric distribution to provide “noise-free” differential privacy, where the randomness of quantization itself satisfies DP requirements [41].

2.2.2 Layer-wise and Feature-Adaptive Noise Uniform noise injection treats all model parameters equally, which is suboptimal given the heterogeneous sensitivity of neural network layers. **LaDP-FL** introduces layer-wise differential privacy, identifying critical layers via L_2 norm thresholds and quantifying leakage using KL divergence. It adaptively scales noise inversely to layer distribution divergence, achieving significant accuracy improvements over full-DP baselines [42]. **Adaptive Differential Privacy (ADP)** leverages Grad-CAM to compute importance weights for convolutional layers, injecting noise primarily into low-importance weights to preserve salient features and interpretability [43]. In the domain of speech recognition, **Sotto Voce** implements

instance-level gradient reconstruction and clamping, tailoring the noise to the specific acoustic model architecture to minimize utility loss [44].

2.2.3 Objective Perturbation vs. Output Perturbation While output perturbation (adding noise to gradients or models) is common, **objective perturbation** offers a theoretically superior alternative for convex problems. **DP-IADMM** integrates objective perturbation into the Inexact ADMM framework, injecting Laplace noise directly into the local subproblem objective functions. This approach randomizes the optimization landscape rather than just the output, yielding superior utility-privacy trade-offs compared to output perturbation methods, particularly in settings with multiple local updates [45], [46].

2.3 Communication-Efficient and Compressed DP-FL

Communication bandwidth is a primary bottleneck in FL, and DP noise exacerbates this by increasing the variance of updates. Recent works focus on joint optimization of compression and privacy.

2.3.1 Sparsification and Sketching **Gradient sparsification** reduces the dimensionality of updates, which inherently lowers sensitivity and allows for less noise injection. **Fed-SPA** integrates random sparsification with Gaussian perturbation, demonstrating that sparsifying updates before perturbation amplifies privacy guarantees via subsampling theorems, saving up to 86% in communication costs while maintaining accuracy [47]. **DP-FCRN** extends this to second-order optimization, using random- k sparsification in a Cubic Regularized Newton method to reduce sensitivity by a factor of $\sqrt{k/d}$, enabling lower noise intensity [48].

Advanced compression techniques utilize **sketching**. **DPSFL** compresses gradients via Count Sketch and injects Gaussian noise directly into the sketch counters. This reduces the required noise magnitude proportional to the dimension reduction, outperforming standard DP-FL methods [49]. **DOMe** employs correlation-aware sketching, projecting high-dimensional gradients into a low-dimensional subspace using streaming PCA before applying clipping and noise, leveraging the temporal correlation of gradient updates [50].

2.3.2 Quantization as a Privacy Mechanism Quantization is increasingly viewed not just as a compression tool but as a privacy mechanism. **Randomized Quantization Mechanism (RQM)** augments the input range and sub-samples quantization levels, providing sufficient entropy for DP without explicit additive noise, thus reducing variance [51]. **CorBin-FL** utilizes correlated binary stochastic quantization where paired clients share common randomness to cancel out quantization errors during aggregation, significantly reducing Mean Squared Error (MSE) compared to independent quantization [52]. **BQ-SGD** unifies quantization error and binomial noise into a single perturbation vector, theoretically deriving the trade-offs between bit-width and privacy budget [18].

2.4 Adaptive Optimization and Dynamic Privacy Budgeting

Static privacy budgets often lead to inefficient resource usage, as early training rounds may require less privacy protection than later stages where models converge.

2.4.1 Time-Adaptive and Geometric Noise Scheduling **Spend-as-You-Go** partitions training into “saving” and “spending” phases, utilizing reduced sampling rates in early rounds to accumulate privacy budget for later, more critical stages [53]. **Amplitude-Varying Perturbation (AV-DP)** implements a geometric series for noise variance, allowing higher noise in early rounds for rapid convergence and reducing it later, or vice versa depending on the architecture, analytically deriving the optimal aggregation rounds [54]. **Dynamic Privacy Allocation** employs a primal-dual algorithm with geometrically decreasing noise variance, stabilizing convergence by allocating higher variance early when noise contributes less to final error [55].

2.4.2 Client Selection and Sampling Strategic client selection can amplify privacy and utility. **IDP-FedAvg** translates the centralized SAMPLE algorithm to FL, computing client-specific sampling rates based on heterogeneous privacy budgets. This allows clients with looser privacy constraints to participate more frequently, outperforming uniform noise scaling [56]. **Privacy-Aware Client Selection** formulates client selection as a convex optimization problem, balancing dataset size against noise variance to minimize the sum of selection gap and privacy-induced error [57]. In wireless settings, **OptSched** jointly optimizes user scheduling and DP noise standard deviations, prioritizing users with better channels and larger data chunks to improve convergence [28].

2.5 Hybrid Frameworks: Cryptography and DP

Combining DP with cryptographic primitives like Homomorphic Encryption (HE) and Secure Multi-Party Computation (MPC) provides defense-in-depth, protecting against both passive inference and active collusion.

2.5.1 DP with Secure Aggregation **Secure Aggregation (SecAgg)** is often paired with DP to hide individual updates. **DDP-SA** integrates client-side LDP with additive secret sharing across intermediate servers, ensuring that even if the server is compromised, the privacy guarantees hold via post-processing invariance [33]. **PRECAD** employs a two-server model where clients split updates into shares; the servers perform crypto-aided validation of clipping norms and inject noise server-side, simulating Centralized DP without a trusted single party [58]. **Gopa** utilizes pairwise-correlated Gaussian noise exchange over random graphs to mask values, achieving trusted curator utility with only $O(\log n)$ communication and verifiable correctness via zero-knowledge proofs [59].

2.5.2 DP with Homomorphic Encryption and MPC **HE-DP** frameworks combine HE for secure aggregation with DP for output perturbation. **Poisson quantisation** is introduced in [20] as an operator that commutes with summation, preserving DP guarantees while operating within the bounded plaintext modulus of HE schemes. **SPEED** integrates distributed DP with Fully Homomorphic Encryption (FHE), utilizing a non-interactive homomorphic argmax operator via TFHE bootstrapping to aggregate noisy teacher predictions securely [60]. In vertical FL, **HDP-VFL** replaces heavy HE operations with Gaussian noise injection into intermediate results, achieving differential privacy with negligible runtime overhead compared to pure cryptographic solutions [61]. **MPC+DP** protocols simulate a trusted curator via MPC, generating secret-shared Laplace noise entirely within the protocol to apply global DP noise only once to the final model [62].

2.6 Decentralized and Peer-to-Peer DP-FL

Removing the central server entirely introduces new challenges for privacy accounting and aggregation.

2.6.1 Gossip and Random Walk Protocols In decentralized topologies, **Network DP** defines privacy relative to a user’s local view of message transcripts. Token-walk protocols, where a state vector traverses the network accumulating noisy contributions, provide $O(1/\sqrt{n})$ privacy amplification [63]. **Zip-DL** injects locally correlated Gaussian noise that sums to zero over a node’s neighborhood, neutralizing the impact on global convergence while maintaining privacy against honest-but-curious neighbors [64]. **DECOR** utilizes pairwise-canceling correlated noise between neighbors in gossip averaging, matching Central DP trade-offs with significantly lower loss than LDP [65].

2.6.2 Blockchain and Smart Contracts Blockchain offers an immutable ledger for decentralized aggregation. **PT-DL** utilizes Ethereum smart contracts to orchestrate agent interactions, employing Zero-Knowledge Proofs to filter malicious updates while maintaining DP guarantees [66]. **DP-BBVFL** replaces the central server with smart contracts for vertical FL, using the Poisson Binomial Mechanism to inject calibrated noise compatible with blockchain integer arithmetic [67]. **BEAS** implements gradient pruning on a permissioned blockchain to enhance privacy-utility trade-offs, leveraging the immutability of the chain to prevent tampering [68].

2.7 Specialized Domains and Applications

2.7.1 Healthcare and Medical Imaging Medical data poses unique challenges due to its sensitivity and heterogeneity. **Dopamine** coordinates noise scales via secure aggregation across hospitals to achieve central DP bounds with reduced individual noise variance [69]. **DPResNet** modifies ResNet architectures by replacing BatchNormalization with GroupNormalization to ensure compatibility with DP, overcoming the dependency on batch statistics that breaks privacy guarantees [70]. **FedLAP-DP** shares differentially private loss approximations (synthetic data) instead of biased local updates, enabling unbiased global optimization on non-IID medical data [71]. **Adaptive Intermediary** strategies split clients into sub-clients to increase the effective participant count, reducing noise magnitude while preserving DP guarantees in cross-silo medical imaging [72].

2.7.2 Wireless and Over-the-Air Computation Wireless channels offer inherent noise that can be harnessed for privacy. **AirFL-DP** leverages channel noise in multi-antenna MAC fading channels as a natural privacy mechanism, optimizing beamforming to achieve “free” DP without artificial noise injection [29]. **DP-AirComp** harnesses receiver noise perturbation via power control, analytically deriving the SNR-privacy trade-off [73]. **RIS-Enabled DP-FL** utilizes Reconfigurable Intelligent Surfaces to reconfigure channel propagation, boosting signal power to counteract privacy-induced noise without violating power constraints [74].

2.7.3 Recommendation Systems and NLP In recommendation systems, **PrivateRec** decomposes high-dimensional user embeddings into low-dimensional attention coefficients, perturbing only the coefficients to preserve utility [75]. **BGTplanner** uses Gaussian Process Regression to predict training rewards and dynamically allocate privacy budgets in federated recommenders, improving RMSE significantly [76]. For NLP, **FedMental** evaluates DP-FL for mental health detection, noting that while standard FL matches centralized performance, DP introduces catastrophic utility loss in sparse linguistic contexts, highlighting the need for careful tuning [77]. **Sotto Voce** adapts DP-SGD hooks for federated acoustic model training, demonstrating feasibility in speech recognition despite high privacy budget requirements [44].

2.8 Fairness, Robustness, and Security

2.8.1 Privacy-Fairness Trade-offs DP mechanisms can exacerbate fairness disparities by disproportionately affecting underrepresented groups. **FedFDP** integrates a fairness regularization term into the local loss function and employs a fair-clipping strategy to balance sensitivity control and fairness adjustment [21]. **PUFFLE** dynamically adjusts fairness regularization weights based on target disparity thresholds, enabling precise control over the privacy-fairness-utility trilemma [78]. Conversely, **FedPF** models the interaction as a zero-sum game, proving an inverse relationship where stricter DP fundamentally limits bias detection capabilities [79].

2.8.2 Byzantine Robustness Malicious clients can exploit DP noise to hide poisoning attacks. **DP-BREM** adds noise to aggregated client momentum rather than local gradients, utilizing centered clipping for Byzantine robustness while preserving the signal-to-noise ratio [24]. **SecureDL** integrates additive secret sharing with cosine similarity-based filtering to neutralize scaling attacks in a decentralized setting [23]. **DeSMP** highlights the vulnerability of DP-FL, demonstrating how adversaries can optimize attack distributions to hide malicious updates within the statistical bounds of DP noise [80].

2.8.3 Empirical Privacy Auditing Theoretical bounds often overestimate privacy. **CANIFE** crafts orthogonal canary samples to empirically measure privacy leakage, revealing gaps between theoretical ϵ and empirical reality [81]. **Empirical LDP Privacy Tests** employ hypothesis testing frameworks with specific adversary instantiations to validate LDP bounds, showing that white-box adversaries can achieve theoretical limits while black-box observers cannot [82].

3. Challenges and Outlook

Despite significant progress, the field of Privacy-Preserving Federated Learning with Differential Privacy faces several critical challenges that hinder widespread deployment.

3.1 The Utility-Privacy-Communication Trilemma

The fundamental tension between privacy, utility, and communication efficiency remains unresolved. While techniques like sparsification [47] and quantization [18] mitigate communication costs, they often introduce additional bias or variance that compounds with DP noise. Recent works indicate that achieving strong privacy ($\epsilon < 1$) in deep learning tasks often leads to unacceptable accuracy drops, particularly in non-IID settings [9], [77]. The “cost of privacy” is not uniform; it scales inversely with dataset size and is exacerbated by data heterogeneity [83], [57]. Future research must focus on **adaptive mechanisms** that dynamically balance these three objectives based on real-time convergence metrics and data characteristics, moving beyond static hyperparameters [84], [53].

3.2 Vulnerabilities in Secure Aggregation and Trust Models

The reliance on Secure Aggregation (SecAgg) to amplify privacy is increasingly scrutinized. **Sybil attacks** pose a severe threat: as demonstrated in [5], a malicious server can inject sybil devices to isolate a target user’s gradient, effectively bypassing SecAgg and reducing Distributed DP to weak Local DP. Furthermore, the **Shuffle Model** relies on the non-collusion assumption between the shuffler and analyzer; if this assumption fails, privacy guarantees collapse to LDP levels [13], [85]. Developing **robust aggregation protocols** that can detect and mitigate sybil injections without

compromising privacy, and designing **trust-minimized shufflers** (e.g., via MPC or TEEs), are urgent priorities.

3.3 Computational and System Overhead

Hybrid approaches combining DP with HE or MPC offer strong security but incur prohibitive computational overheads. For instance, HE-based aggregation can be orders of magnitude slower than plaintext aggregation, rendering it impractical for real-time applications on edge devices [19], [86]. Similarly, complex adaptive clipping and layer-wise noise injection mechanisms increase client-side computation, straining resource-constrained IoT devices [42], [15]. Research into **lightweight cryptographic primitives** and **hardware-accelerated DP mechanisms** (e.g., leveraging GPU/TPU features for noise generation) is essential to bridge the gap between theoretical feasibility and practical deployment.

3.4 Privacy Accounting and Heterogeneity

Accurate privacy accounting in heterogeneous FL settings remains an open problem. Existing composition theorems often assume uniform sampling and homogeneous privacy budgets, which rarely hold in practice. **Heterogeneous DP** scenarios, where clients have varying privacy requirements, complicate the aggregation process and privacy budget tracking [87], [56]. Moreover, **client dropout** and asynchronous updates violate the uniform sampling assumptions required for many privacy amplification proofs [10], [88]. Developing **tighter, adaptive privacy accountants** that can handle dynamic participation, heterogeneous budgets, and non-IID data distributions is critical for providing reliable guarantees.

3.5 Empirical Validation and Benchmarking

There is a disconnect between theoretical privacy bounds and empirical leakage. As shown by **CANIFE** and other auditing tools, empirical ϵ can be significantly lower (better) or higher (worse) than theoretical predictions depending on the attack model [81], [82]. The lack of standardized benchmarks for evaluating DP-FL under realistic attack scenarios hinders progress. Initiatives like **PrivacyBench** aim to quantify non-additive interaction effects of hybrid techniques, but more comprehensive benchmarks covering diverse domains (healthcare, finance, NLP) and attack vectors (reconstruction, membership, property inference) are needed [89], [90].

3.6 Future Directions

Future research should explore **Personalized DP-FL**, where privacy guarantees are tailored to individual user preferences and data sensitivity, potentially using meta-learning or hypernetworks [91], [92]. The integration of **Generative AI** for synthetic data generation under DP constraints offers a promising avenue to decouple utility from privacy noise, allowing models to train on high-fidelity synthetic datasets [6], [93]. Additionally, the emergence of **Quantum Federated Learning** introduces new opportunities and challenges, with initial works exploring DP in hybrid quantum-classical models [94], [95]. Finally, **regulatory alignment** is crucial; frameworks that map technical DP parameters to legal compliance requirements (e.g., GDPR) will facilitate adoption in regulated industries [96], [97].

4. Conclusion

The convergence of Federated Learning and Differential Privacy represents a cornerstone of modern privacy-preserving machine learning. This survey has synthesized a vast body of literature, revealing a field that has matured from simple noise injection to sophisticated, adaptive, and hybrid architectures. We have categorized methodologies based on trust models, highlighting the evolution from Local DP to Distributed and Shuffle models that leverage secure aggregation for privacy amplification. We examined innovations in noise mechanisms, including adaptive clipping, layer-wise perturbation, and quantization-based privacy, which collectively push the boundaries of the privacy-utility trade-off.

However, significant challenges remain. The vulnerability of secure aggregation to active attacks, the computational burden of cryptographic hybrids, and the complexity of privacy accounting in heterogeneous, dynamic environments necessitate continued research. The path forward lies in developing robust, efficient, and empirically validated frameworks that can operate under realistic threat models. As the demand for collaborative AI grows across sensitive domains like healthcare and finance, the techniques surveyed herein will play a pivotal role in ensuring that the benefits of machine learning are realized without compromising individual privacy. The integration of DP into FL is not merely a technical add-on but a fundamental requirement for trustworthy, decentralized intelligence.

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